

Peripheral opioid receptor antagonism alleviates fentanyl-induced cardiorespiratory depression and is devoid of aversive behavior

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eLife Assessment

This manuscript represents a **fundamental** contribution demonstrating that fentanyl-induced respiratory depression can be reversed with a peripherally-restricted mu opioid receptor antagonist. The paper reports **compelling** and rigorous physiological, pharmacokinetic, and behavioral evidence supporting this major claim, and furthers mechanistic understanding of how peripheral opioid receptors contribute to respiratory depression. These findings reshape our understanding of opioid-related effects on respiration and have significant therapeutic implications given that medications currently used to reverse opioid overdose (such as naloxone) produce severe aversive and withdrawal effects via actions within the central nervous system.

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Abstract

Millions of Americans suffering from Opioid Use Disorders (OUD) face a high risk of fatal overdose due to opioid-induced respiratory depression (OIRD). Fentanyl, a powerful synthetic opioid, is a major contributor to the rising rates of overdose deaths. Reversing fentanyl overdoses has proved challenging due to its high potency and the rapid onset of OIRD. We assessed the contributions of central and peripheral mu opioid receptors (MORs) in mediating fentanyl-induced physiological responses. The peripherally restricted MOR antagonist naloxone methiodide (NLXM) both prevented and reversed OIRD to a degree comparable to that of naloxone (NLX), indicating substantial involvement of peripheral MORs to OIRD. Interestingly, NLXM-mediated OIRD reversal did not produce aversive behaviors observed after NLX. We show that neurons in the nucleus of the solitary tract (nTS), the first central synapse of peripheral afferents, exhibit a biphasic activity profile following fentanyl exposure. NLXM pretreatment attenuates this activity, suggesting that these responses are

mediated by peripheral MORs. Together, these findings establish a critical role for peripheral MORs, including ascending inputs to the nTS, as sites of dysfunction during OIRD. Furthermore, selective peripheral MOR antagonism could be a promising therapeutic strategy for managing OIRD by sparing CNS-driven acute opioid-associated withdrawal and aversion observed after NLX.

Significance Statement

In this study, we compare the central versus peripheral components underlying fentanyl-induced cardiorespiratory depression to prevent overdose deaths. Our data indicate that these effects are, at least partially, due to the activation of mu opioid receptors present in peripheral sites. These findings provide insight into peripheral contributions to fentanyl-induced overdoses and could potentially lead to the development of treatments selectively targeting the peripheral system, sparing individuals from the CNS-driven acute opioid withdrawal generally observed with the use of naloxone.

Introduction

Millions of Americans live with Opioid Use Disorders (OUD) and face a high risk of opioid-induced respiratory depression (OIRD), the leading cause of opioid related deaths (1). The number of fatalities attributed to OIRD has been exacerbated by the rise of distribution and use of synthetic opioids, such as fentanyl (2–5). Fentanyl is a highly lipophilic opioid that readily crosses the blood-brain barrier and binds tightly to mu opioid receptors (MORs), which are abundant in the respiratory centers of the brainstem (1, 6, 7). Compared to other opioids such as heroin and morphine, fentanyl exhibits both a faster onset of OIRD and higher potency for MOR binding (8, 9), minimizing successful prevention of lethal outcomes using naloxone (NLX), a competitive and preferential MOR antagonist (10, 11). Despite its high efficacy in reversing OIRD, NLX also precipitates unpleasant withdrawal symptoms and aversion, as reported in patients and preclinical models (11–14), making its implementation for managing OUD challenging. Therefore, a better understanding of the pharmacology, specific brain regions and pathways, and physiological responses induced by high doses of fentanyl are necessary to develop effective prevention and intervention strategies, ultimately saving lives of patients suffering from OUD.

Several regions have been implicated as key sites of triggering OIRD, including the Pre-Botzinger Complex in the ventral medulla and the parabrachial nucleus and Kölliker-Fuse nucleus in the pons (1, 15–20). Opioids act directly within these networks to alter inspiratory and expiratory phase duration, leading to overall reductions in respiratory rate and apneas (18, 20, 21). In addition, the nucleus of the solitary tract (nTS), located in the dorsal brainstem, is the first central site that receives sensory afferent information via cranial nerves related to cardiorespiratory, gustatory and gastrointestinal function (22). Acute hypoxia robustly activates nTS neurons (23, 24) to initiate appropriate autonomic and cardiorespiratory responses that facilitate a return to homeostatic physiological states. MORs are expressed throughout the entire caudal-rostral extent of the nTS, where they are located postsynaptically on nTS neurons and on vagal afferent fibers that terminate within the nTS (25–27). Opioids induce both systemic and brain hypoxia (28, 29) and produce an increase in Fos-immunoreactivity in the nTS (26, 30). However, engaging MOR signaling in the nTS impairs hypoxic ventilatory responses (7). This suggests that OIRD may result from nTS dysfunction failing to engage appropriate cardiorespiratory responses, thereby leading to prolonged OIRD. However, the specific contributions of nTS MOR, including ascending MOR-expressing inputs from the periphery to this region, in driving aberrant cardiorespiratory depression are still not fully understood. In addition

to their high expression throughout brainstem respiratory nuclei, MORs are also expressed in the periphery, including sensory ganglia, lung afferents, cardiac tissue and cranial nerves that terminate within the nTS (31 [↗](#), 32 [↗](#)). Naloxone Methiodide (NLXM) is a quaternary derivative of NLX that does not cross the blood brain barrier at low doses, making it a useful tool to assess the peripheral vs central contribution of MORs in opioid-induced physiology and behavior (10 [↗](#), 33 [↗](#)). Despite its lower binding affinity for MOR (10 [↗](#)), NLXM has been shown to reverse opioid-induced hypoventilation and brain hypoxia in rodent models (12 [↗](#), 33 [↗](#), 34 [↗](#)), suggesting that peripheral opioid receptors may play a greater role in OIRD than previously thought. However, no studies to date have provided a comprehensive assessment of opioid-induced cardiorespiratory depression and systemic hypoxia combined with a real-time assessment of MOR-mediated activity of nTS neurons.

In the present study, we examined the relative contributions of central and peripheral MORs mediating fentanyl-induced depression of cardiorespiratory parameters and investigated potential mechanisms of opioid-induced dysfunction within the nTS. We report that fentanyl-induced cardiorespiratory depression and prolonged systemic hypoxia can be prevented or reversed by NLXM to a comparable degree as NLX. As compared to NLX, NLXM-mediated reversal of OIRD did not produce aversive-like behaviors. Fentanyl induced robust Fos-IR expression in the nTS in a dose-dependent manner, and this activation can be attributed to a combination of direct effects of opioids acting at MOR located within the nTS and the subsequent hypoxia that develops after fentanyl exposure. Given that the majority of nTS MOR is located on vagal afferent fibers, we utilized various techniques to evaluate opioid-induced activity within the nTS and manipulate nTS MOR signaling during OIRD. Using wireless *in vivo* fiber photometry, we show that fentanyl evokes a biphasic activity profile in nTS neurons, characterized by a transient increase followed by a prolonged decrease in neuronal excitability. NLXM pretreatment strongly attenuated this biphasic response, indicating that fentanyl-induced nTS activity is influenced by MOR-expressing peripheral afferents. Together, these findings provide novel insights into peripheral mechanisms mediating OIRD and support peripheral MOR antagonism as a promising therapeutic strategy for managing OIRD for patients suffering from OUD while sparing the CNS-driven acute aversive behaviors that are observed with the use of NLX.

Results

Intravenous fentanyl induces respiratory depression and activates nTS neurons in a dose-dependent manner

We first examined the effect of intravenous fentanyl to induce cardiorespiratory depression and activate neurons in the nucleus tractus solitarius (nTS). Catheterized rats received intravenous saline (1 ml/kg) or fentanyl at a range of doses (2, 20, or 50 $\mu\text{g/kg}$; **Figure 1A** [↗](#)), and cardiorespiratory parameters were measured up to 60 minutes. The first 20 minutes of these responses are shown in **Figure 1B-D** [↗](#). As shown in **Table 1** [↗](#), baseline cardiorespiratory parameters were similar between male and female rats that received 20 $\mu\text{g/kg}$ fentanyl. There was a small but significant difference in basal heart rate in female rats that received 50 $\mu\text{g/kg}$ fentanyl (**Table 2** [↗](#)). Neither saline nor 2 $\mu\text{g/kg}$ fentanyl evoked a significant change in cardiorespiratory parameters. In comparison, both 20 and 50 $\mu\text{g/kg}$ fentanyl produced rapid decreases in oxygen saturation, heart rate, and respiratory rate, with the largest decrease occurring after 50 $\mu\text{g/kg}$ fentanyl. The nadir (defined as the lowest point measured after fentanyl) and recovery (calculated as 90% of baseline values prior to fentanyl) for each physiological parameter were assessed in all groups (**Figure 1E-J** [↗](#)). There was no significant difference in the nadir or recovery values between saline- and 2 $\mu\text{g/kg}$ fentanyl-treated rats. In comparison, rats given 50 $\mu\text{g/kg}$ fentanyl exhibited a significantly lower nadir in oxygen saturation (**Figure 1E** [↗](#)) and respiratory rate (**Figure 1I** [↗](#)) compared to 20 $\mu\text{g/kg}$. In addition, the recovery time was significantly longer in rats that received 50 $\mu\text{g/kg}$ fentanyl compared to 20 $\mu\text{g/kg}$ fentanyl (**Figure 1F,J** [↗](#)). The nadir of

bradycardia was similar between 20- and 50 $\mu\text{g/kg}$ rats (**Figure 1G**), although the duration of the response was significantly longer in rats that received the highest dose of fentanyl (**Figure 1H**). Overall, we observed no sex differences in the nadir or recovery time in rats that received 20 $\mu\text{g/kg}$ fentanyl. Female rats that received 50 $\mu\text{g/kg}$ fentanyl exhibited a significantly longer recovery of respiratory rate, although oxygen saturation and heart rate recover times were similar to males (**Tables 3,4**). We next examined nTS activation (Fos-IR) following intravenous fentanyl (**Figure 1K**). A significant increase in Fos-IR was observed after 2 $\mu\text{g/kg}$ fentanyl, (**Figure 1L**), a dose that did not induce cardiorespiratory depression. Further dose-dependent increases in Fos-IR were observed after 20 and 50 $\mu\text{g/kg}$ fentanyl. We evaluated a subpopulation of catecholaminergic nTS neurons, which are critical for full expression of hypoxia-evoked cardiorespiratory responses (23, 35) for activation after intravenous fentanyl (**Figure 1M**). These cells exhibited increased Fos-IR after 2 $\mu\text{g/kg}$ fentanyl and were further activated at the highest dose of fentanyl (50 $\mu\text{g/kg}$).

Peripheral opioid receptor antagonism prevents fentanyl-induced respiratory depression

The above data provide a comprehensive assessment of the dose-dependent changes in cardiorespiratory parameters and nTS neuronal activation evoked by intravenous fentanyl. To gain insight into the extent to which central and peripheral opioid receptors mediate these effects, separate cohorts of rats received intravenous administration of the centrally and peripherally acting competitive opioid receptor antagonist naloxone (NLX; 1 mg/kg) or the peripherally restricted opioid receptor antagonist naloxone methiodide (NLXM, 1 and 5 mg/kg). Initial biodistribution experiments were performed to confirm the peripheral restriction of NLXM. Catheterized rats that received intravenous NLXM (1 or 5 mg/kg) or NLX (1 mg/kg) were evaluated for detection of NLX or NLXM in brain and plasma samples collected either 2 or 10 minutes after administration (**Figure 2A**). Neither dose of NLXM was detected within the central nervous system at either time point evaluated (**Figure 2B,C**). Importantly, NLX was not detected in these NLXM-treated groups. In comparison, rats that received intravenous NLX exhibited significant, time-dependent detection of NLX in both plasma and brain samples (**Figure 2D**). Thus our data are in agreement with previous reports confirming the peripheral restriction of NLXM (33, 36). We next evaluated cardiorespiratory changes from baseline in rats that received intravenous NLX or NLXM at either dose (**Figure S3A**). Data indicate that neither NLX nor NLXM significantly altered cardiorespiratory parameters from their baseline parameters (**Figure S3E-J**). Based on these findings, we utilized NLXM to further investigate the roles of peripheral MOR antagonism in the context of evaluating OIRD.

Catheterized rats received intravenous NLXM (1 or 5 mg/kg) or NLX (1 mg/kg) prior to intravenous fentanyl administration at 20 $\mu\text{g/kg}$ (**Figure 3A**) or 50 $\mu\text{g/kg}$ (**Figure S1A**). As shown in **Figure 3B-D** and **Figure S1B-D**, the responses evoked by 20 $\mu\text{g/kg}$ and 50 $\mu\text{g/kg}$ fentanyl induced an onset and duration of cardiorespiratory depression that was similar to what was observed in **Figure 1**. As expected, NLX pretreatment completely blocked the decrease in cardiorespiratory parameters at both 20 $\mu\text{g/kg}$ and 50 $\mu\text{g/kg}$ doses of fentanyl. In rats that received 20 $\mu\text{g/kg}$ fentanyl, 1 mg/kg NLXM pretreatment prevented bradycardic and hypoventilatory responses to fentanyl (**Figure 3B-D**). For all parameters, saline-pretreated rats exhibited lower nadir values than NLX- and NLXM-pretreated rats, and there was no significant difference in the nadir or duration between NLX (1 mg/kg) and NLXM (1 mg/kg) pretreated rats that received 20 $\mu\text{g/kg}$ fentanyl (**Figure 3E-J**). Although there was a small drop in the nadir of oxygen saturation in NLXM-pretreated rats, this was not significantly different than NLX-pretreated animals (**Figure 3E**, $p=0.2292$). In rats that received 50 $\mu\text{g/kg}$ fentanyl, 1 mg/kg NLXM pretreatment was unable to prevent the initial decrease in oxygen saturation and heart rate (**Figure S1E,G**). However, the duration of these responses was significantly shorter compared to saline-treated rats (**Figure S1F,H,J**). A higher dose of NLXM (5 mg/kg), which does not cross the blood brain barrier (**Figure**

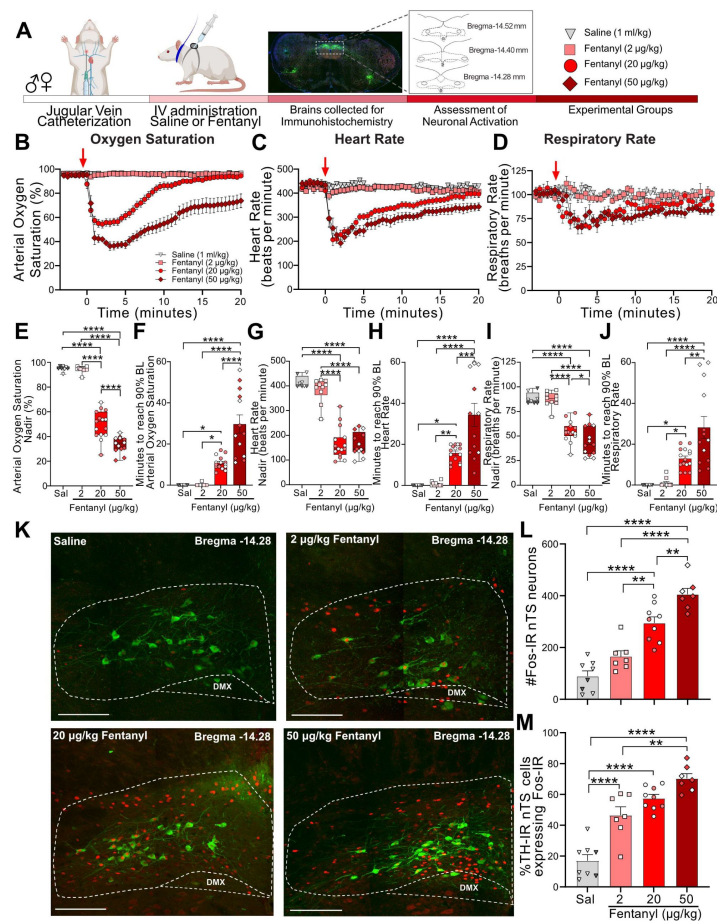


Figure 1

Fentanyl induces cardiorespiratory depression and nTS neuronal activation in a dose-dependent manner

A. Schematic showing the timeline for OIRD collar experiments. Male and Female SD rats underwent jugular catheterization surgery. One week later, conscious rats in their home cages were fitted with a pulse oximeter collar and received intravenous administration of either saline (n=8, triangles) or fentanyl at doses of 2 µg/kg (n=10, squares), 20 µg/kg (n=15, circles), or 50 µg/kg (n=13, diamonds). In all graphs, males are represented by filled symbols, and females by open symbols. Baseline cardiorespiratory parameters were measured for 3 minutes prior to intravenous administration and up to 60 minutes after saline or fentanyl. At the conclusion of experiments, brains were collected from a subset of animals in each group and processed for Fos- and TH-immunoreactivity. Time course data showing oxygen saturation (**B**), heart rate (**C**) and respiratory rate (**D**) data measurements in all groups before and after saline or fentanyl. Changes in oxygen saturation were assessed as nadir (lowest value; **E,G,I**) and time to recover to 90% of baseline (pre-saline/fentanyl; **F,H,J**) administration. There was a dose-dependent decrease in the nadir for oxygen saturation (One-way ANOVA, $F_{(3,42)} = 186.3$, $p < 0.0001$; Tukey's post-hoc **** $p < 0.0001$), heart rate (One-way ANOVA $F_{(3,42)} = 74.93$, $p < 0.0001$; Tukey's post-hoc **** $p < 0.0001$) and respiratory rate (One-way ANOVA, $F_{(3,42)} = 45.60$, $p < 0.0001$; Tukey's post-hoc **** $p < 0.0001$, * $p < 0.05$). Time to recover to 90% baseline Oxygen Saturation (One-way ANOVA, $F_{(3,42)} = 28.72$, $p < 0.0001$; Tukey's post-hoc **** $p < 0.0001$, * $p < 0.05$) heart rate (One-way ANOVA, $F_{(3,42)} = 23.89$, $p < 0.0001$; Tukey's post-hoc **** $p < 0.0001$, *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$) and respiratory rate (One-way ANOVA, $F_{(3,42)} = 16.53$, $p < 0.0001$; Tukey's post-hoc **** $p < 0.0001$, ** $p < 0.01$, * $p < 0.05$). **K**. Merged photomicrographs of coronal brainstem sections displaying representative Fos- and TH-immunoreactivity in the nTS of rats that received intravenous saline or fentanyl. Scale bar = 200 µm. **L**. Mean data show that 20 and 50 µg/kg fentanyl induced a significantly greater number of Fos-IR cells in the nTS compared to saline and 2 µg/kg fentanyl. (One-way ANOVA, $F_{(3,27)} = 37.13$, $p < 0.0001$; Tukey's post hoc **** $p < 0.0001$, ** $p < 0.001$) **M**. Mean data showing the percentage of TH-IR nTS neurons expressing Fos-IR. Rats that received 2 µg/kg fentanyl displayed a significantly higher increase in activated TH-IR neurons compared to saline-treated rats. Both 20 and 50 µg/kg fentanyl induced a significantly higher percentage of activation in TH-IR cells compared to saline and 2 µg/kg fentanyl. (One-way ANOVA, $F_{(3,27)} = 37.13$, $p < 0.0001$; Tukey's post-hoc **** $p < 0.0001$, ** $p < 0.01$)

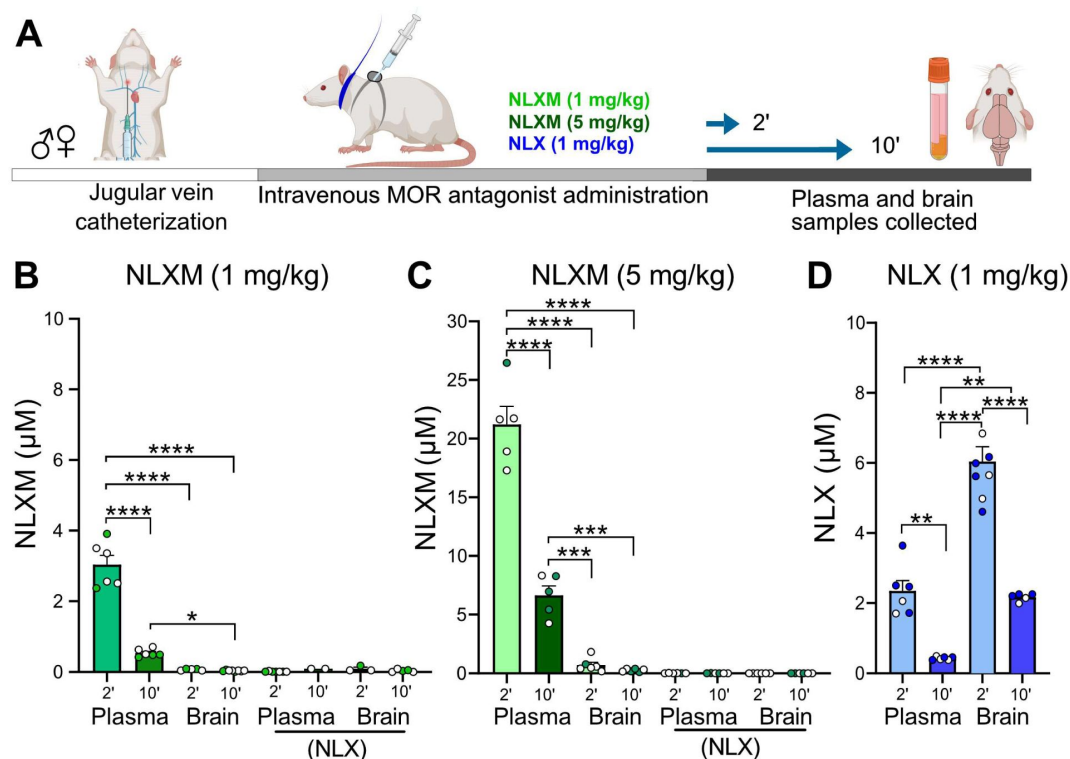


Figure 2

NLXM does not cross the blood brain barrier.

A. Timeline of biodistribution experiments. Male and female SD rats underwent intravenous jugular catheterization surgery. The following week, rats were anesthetized and received intravenous NLXM (1 or 5 mg/kg) or NLX (1 mg/kg). Rats were sacrificed 2 and 10 minutes after intravenous administration of MOR antagonists. Plasma was extracted from trunk blood and intact brains were removed and rapidly frozen with liquid nitrogen. **B.** Amount of NLXM detected in plasma and brain samples at each time point after intravenous NLXM (1 mg/kg). One-way ANOVA, $F_{(3,21)} = 125.6$, $p < 0.0001$; Tukey's post-hoc test, **** $p < 0.0001$; * $p < 0.05$. No NLX formation was detected in these samples. **C.** Amount of NLXM detected in plasma and brain samples at each time point after intravenous NLXM (5 mg/kg). One-way ANOVA, $F_{(3,18)} = 144.0$, $p < 0.0001$; Tukey's post-hoc test, **** $p < 0.0001$; *** $p < 0.001$. No NLX formation was detected in these samples. **D.** Amount of NLX detected in plasma and brain samples at each time point after intravenous NLX (1 mg/kg). One-way ANOVA, $F_{(3,21)} = 66.05$, $p < 0.0001$; Tukey's post-hoc test, **** $p < 0.0001$; ** $p < 0.01$.

2C [↗](#)), was then used in a separate group of animals that received 50 µg/kg fentanyl. Pretreatment with 5 mg/kg NLXM attenuated cardiorespiratory depression. While the nadir of oxygen saturation was significantly lower than NLX-pretreated rats, the duration of this effect was not significantly different between groups (**Figure S1E-J** [↗](#)). No sex differences were observed in the nadir or recovery time of any parameter (**Tables 3,4** [↗](#)). Together, these data demonstrate that peripheral MOR antagonism can sufficiently prevent OIRD by 20 µg/kg fentanyl and strongly attenuates OIRD evoked by 50 µg/kg fentanyl.

Neuronal activation, as measured by Fos-IR, was examined in the nTS of rats pretreated with saline or opioid receptor antagonists followed by 20 µg/kg (**Figure 3K** [↗](#)) or 50 µg/kg fentanyl (**Figure S1K** [↗](#)). Saline pretreatment evoked the highest degree of fentanyl-induced Fos-IR in the nTS, including in a subpopulation of TH-IR neurons. Blocking peripheral MOR signaling with NLXM did not decrease fentanyl-induced nTS neuronal activation, while NLX, which blocks both central and peripheral MORs, resulted in a significant reduction in nTS activation (**Figure 3L,M** [↗](#)). In rats that received the highest dose of fentanyl, the number of nTS cells displaying Fos-IR was similar between rats pretreated with saline and NLXM at both doses, with the only significant reduction occurring in NLX-pretreated animals (**Figure S1L** [↗](#)). This degree of activation was also observed in catecholaminergic cells (**Figure S1M** [↗](#)), although around 30% of TH-IR neurons also displayed Fos-IR in NLX-pretreated rats.

Peripheral opioid receptor antagonism reverses fentanyl-induced respiratory depression

The above data demonstrate that peripheral MOR antagonism prevents fentanyl-induced respiratory depression without affecting nTS neuronal activation as measured by Fos-IR. Given that reversal of OIRD has more translational relevance, we next evaluated the extent to which NLXM reverses cardiorespiratory depression after it has been induced by 20 µg/kg (**Figure 4A** [↗](#)) or 50 µg/kg fentanyl (**Figure S2A** [↗](#)). Both 20 (**Figure 4B-D** [↗](#)) and 50 µg/kg fentanyl (**Figure S2B-D** [↗](#)) induced rapid decreases in oxygen saturation, heart rate and respiratory rate, similar to what was observed in **Figure 1** [↗](#). The fentanyl-induced nadir in all physiological parameters prior to intravenous administration of saline, NLX or NLXM was similar between groups. In rats that received 20 µg/kg fentanyl, both intravenous NLX (1 mg/kg) and NLXM (1 mg/kg) restored cardiorespiratory values to baseline significantly faster than saline-pretreated animals (**Figure 4F,H,J** [↗](#)). This reversal of cardiorespiratory parameters was similar between NLX- and NLXM-treated animals. In rats that received 50 µg/kg fentanyl, NLX rapidly restored all physiological parameters to their baseline values (**Figure S2F,H,J** [↗](#)). While 1 mg/kg NLXM did not sufficiently reverse the OIRD induced by 50 µg/kg fentanyl (**Figure S2F** [↗](#)), a higher dose of 5 mg/kg NLXM fully reversed the fentanyl-induced bradycardia and hypoventilation. The time to restore parameters to baseline was not significantly different than NLX. We did not observe any sex differences in the nadir or recovery time of any parameter (**Tables 3,4** [↗](#)). Thus, blocking peripheral MORs sufficiently reverses OIRD at both moderate and high doses of fentanyl.

Because acute hypoxia activates nTS neurons ([24](#) [↗](#)), we investigated the impact of peripheral and central MOR antagonism on fentanyl-induced nTS neuronal activation (**Figure 4K** [↗](#)). At both doses of fentanyl administered, a similar degree of nTS neuronal activation was observed in all groups regardless of MOR antagonist administered after fentanyl (**Figure 4L,M** [↗](#); **Figure S2L,M** [↗](#)). Together, these data demonstrate that selective antagonism of peripheral opioid receptors sufficiently reverses cardiorespiratory depression induced by both moderate and high doses of fentanyl. Moreover, these data suggest that even a brief period of fentanyl-induced cardiorespiratory depression is sufficient to induce robust activation of nTS neurons, and this subsequent MOR antagonism has no additional effect to increase nTS activation.

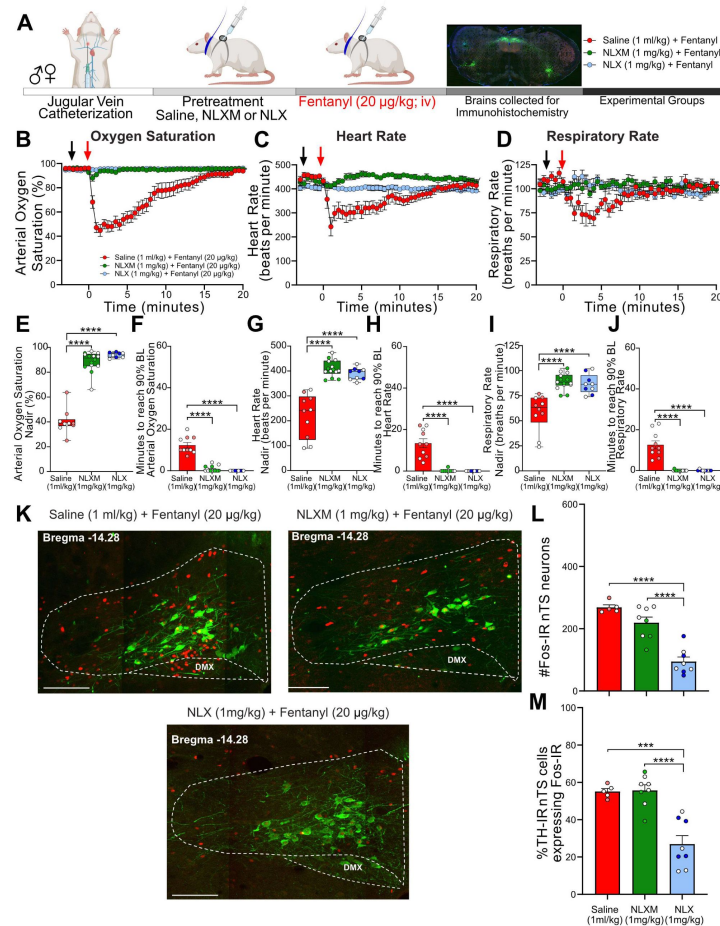


Figure 3.

NLXM attenuates respiratory depression without altering nTS Fos-IR induced by 20 µg/kg fentanyl

A. Schematic showing the timeline for OIRD collar experiments. Male and Female SD rats underwent jugular catheterization surgery. The following week, conscious rats in their home cages received intravenous administration (pretreatment) of either saline (1 ml/kg; n=10), NLX (1 mg/kg; n=9) or NLXM (1 mg/kg; n=15). Two minutes later, all rats received intravenous fentanyl (20 µg/kg). Cardiorespiratory parameters were measured up to 60 minutes after fentanyl. At the conclusion of the experiment, brains were collected from a subset of animals in each group and processed for Fos- and TH-immunoreactivity. Time course data showing oxygen saturation (**B**), heart rate (**C**) and respiratory rate (**D**) data measurements collected in all groups before and after saline, NLX or NLXM pretreatment and after fentanyl. Changes in oxygen saturation were assessed as nadir (lowest value after fentanyl; **E,G,I**) and time to recover to 90% of baseline (**F,H,J**). In all graphs, males are represented by filled symbols, and females by open symbols. Mean nadir values were assessed in saline-, NLX- and NLXM-pretreated rats (Oxygen Saturation: One-way ANOVA, $F_{(2,30)} = 152.0$, $p < 0.0001$; Tukey's post-hoc test, $****p < 0.0001$; Heart Rate: One-way ANOVA, $F_{(2,30)} = 38.58$, $p < 0.0001$; Tukey's post-hoc test, $****p < 0.0001$; Respiratory Rate: One-way ANOVA, $F_{(2,30)} = 20.16$, $p < 0.0001$; Tukey's post-hoc test, $****p < 0.0001$). Time to reach 90% baseline in saline-, NLX- and NLXM-pretreated rats. Mean time to reach 90% baseline was assessed for all parameters. Oxygen Saturation: (One-way ANOVA, $F_{(2,30)} = 81.9$, $p < 0.0001$; Tukey's post-hoc test $****p < 0.0001$; Heart rate: (One-way ANOVA, $F_{(2,30)} = 44.11$, $p < 0.0001$; Tukey's post-hoc test $****p < 0.0001$); Respiratory rate: (One-way ANOVA, $F_{(2,30)} = 46.74$, $p < 0.0001$; Tukey's post-hoc test $****p < 0.0001$).

K. Merged photomicrographs of a coronal brainstem section showing representative Fos- and TH-immunoreactivity in the nTS of rats that received fentanyl with or without saline or NLX/NLXM pretreatment. Scale bar = 200 µm. Fos-IR and the percentage of TH-IR neurons expressing Fos-IR was evaluated in 3 nTS sections per rat. **L.** Mean data show no significant difference in the number of Fos-IR cells between saline- and NLXM-pretreated rats. NLX-pretreated rats displayed the lowest degree of Fos-IR in the nTS. Number of Fos-IR cells: One-way ANOVA, $F_{(2,18)} = 30.52$, $p < 0.0001$; Tukey's post-hoc $****p < 0.0001$. **M.** Mean data of the percentage of TH-IR nTS cells expressing Fos-IR. The percentage of TH-IR cells expressing Fos-IR was similar between saline- and NLXM-pretreated rats. NLX-pretreated rats displayed the lowest percentage of Fos-IR in TH-IR nTS cells. Fos+TH/TH: (One-way ANOVA, $F_{(2,18)} = 21.33$, $p < 0.0001$; Tukey's post-hoc $****p < 0.0001$; $***p < 0.001$).

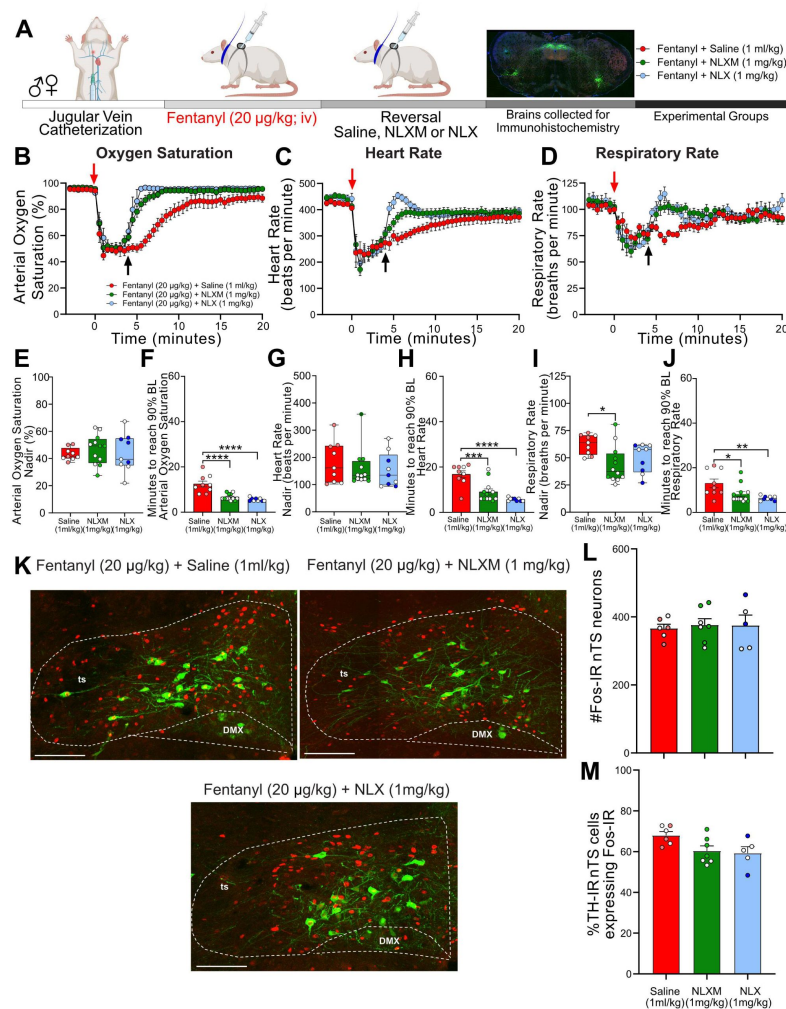


Figure 4.

NLXM-mediated reversal of 20 µg/kg fentanyl-induced respiratory depression is comparable to NLX.

A. Schematic showing the timeline for OIRD reversal experiments. Male and Female SD rats underwent jugular catheterization surgery. The following week, conscious rats in their home cages received intravenous fentanyl (20 µg/kg). Four minutes later, rats received intravenous administration (reversal) of either saline (1 ml/kg; n=9), NLX (1 mg/kg; n=9) or NLXM (1 mg/kg; n=12). Cardiorespiratory parameters were measured up to 60 minutes after fentanyl. At the conclusion of the experiment, brains were collected from a subset of animals in each group and processed for Fos- and TH-immunoreactivity. Time course data showing oxygen saturation (B), heart rate (C) and respiratory rate (D) data measurements collected in all groups before and after fentanyl and saline/antagonist administration. Changes in oxygen saturation were assessed as nadir (lowest value; E, G, I) and time to recover to 90% of baseline (pre-saline/MOR antagonist; F, H, J). In all graphs, males are represented by filled symbols, and females by open symbols. Mean nadir values were assessed in saline-, NLX-, and NLXM-pretreated rats (Oxygen Saturation: One-way ANOVA, $F_{(2,27)} = 0.2916$, $p = 0.7494$; Heart Rate: One-way ANOVA, $F_{(2,27)} = 0.4216$, $p = 0.6602$; Respiratory Rate: One-way ANOVA, $F_{(2,27)} = 4.679$, $p = 0.018$; Tukey's post-hoc test, $*p < 0.05$. Mean time to reach 90% baseline was assessed for all parameters: Oxygen Saturation: One-way ANOVA, $F_{(2,27)} = 22.58$, $p < 0.0001$; Tukey's post-hoc test, $****p < 0.0001$; Heart Rate: One-way ANOVA, $F_{(2,27)} = 19.39$, $p < 0.0001$; Tukey's post-hoc test, $****p < 0.0001$, $***p < 0.001$; Respiratory Rate: One-way ANOVA, $F_{(2,27)} = 6.322$, $p = 0.0001$; Tukey's post-hoc test, $**p < 0.01$, $*p < 0.05$).

K. Merged photomicrographs of coronal brainstem sections displaying representative Fos- and TH-immunoreactivity in the nTS of rats that received fentanyl followed by saline, NLX or NLXM. Scale bar = 200 µm.

L. Mean data of the number of Fos-IR cells evaluated in 3 nTS sections per rat. Number of Fos-IR cells: One-way ANOVA, $F_{(2,15)} = 0.07269$, $p = 0.9302$.

M. Mean data of the percentage of TH-IR cells expressing Fos-IR: One-way ANOVA, $F_{(2,15)} = 3.736$, $p = 0.0482$. There were no significant differences in the number of Fos-IR cells or the percentage of TH-IR cells expressing Fos-IR between groups.

Reversal of Fentanyl-Induced Respiratory Depression with NLXM is not aversive

NLX rapidly reverses OIRD, but produces unwanted side effects including the immediate onset of severe withdrawal (12, 13). The intense symptoms induced during NLX-precipitated withdrawal present additional challenges in managing OUD in patients. Given that peripheral MOR antagonism sufficiently reverses fentanyl-induced cardiorespiratory depression to a degree that is comparable to NLX, we used a conditioned place aversion (CPA) approach to assess whether selective peripheral antagonism of MOR signaling produces aversive behaviors. Rats underwent a 3-session CPA challenge that consisted of pre-conditioning test, conditioning test, and post-conditioning test (Figure 5A). Horizontal locomotion in both compartments of the CPA boxes was assessed during both pre- and post-conditioning tests (Figure 5B). During the pre-conditioning test, rats exhibited no natural preference or aversion for either compartment.

On the conditioning day (Day 2), a divider was placed in the middle of the CPA box. Rats received intravenous fentanyl (50 µg/kg) in their home cage. Four minutes later, rats were placed into the paired compartment of the CPA box and immediately given intravenous saline (1 ml/kg), NLX (1 mg/kg) or NLXM (1 or 5 mg/kg). During the post-conditioning test, the divider was removed, and rats were allowed to explore both compartments of the CPA box. Figure 5C shows the time spent on the conditioned side of the CPA box for all groups. During the post-conditioning test, NLX-treated animals spent significantly less time in the NLX-paired compartment, suggesting that the reversal of OIRD on the conditioning day was aversive. This conditioned place aversion behavior was not observed in rats that received NLXM at either dose (Figure 5D). Importantly, a single conditioning session with fentanyl administered in the home cage followed by saline in the compartment instead of a MOR antagonist did not produce a preference or aversion to the saline-paired compartment. To rule out any potential effects of endogenous opioids influencing the behavioral responses to opioid receptor antagonism, we performed an additional test in which rats were given saline in their home cage, followed by NLX or NLXM in one of the CPA compartments (Figure 5E,F). Neither antagonist produced a preference nor aversion to the MOR antagonist-paired compartment (Figure 5G). There was no difference in CPA scores between these groups (Figure 5H). These findings are consistent with our data demonstrating that NLX and NLXM alone do not alter physiological parameters from baseline (Figure S3).

Fentanyl induces a biphasic activity profile in nTS neurons

As shown in Figure 1, fentanyl induced robust Fos-IR expression in the nTS. While Fos-IR quantification provides insight into the anatomical activation of brainstem subregions in response to fentanyl administration and OIRD reversal, this approach does not provide any temporal resolution of the activity of these cells. Therefore, we used wireless *in vivo* fiber photometry to assess dynamic changes in nTS calcium transient activity after intravenous fentanyl administration in anesthetized rats (Figure 6A). Catheterized rats underwent bilateral injections of an AAV expressing a calcium sensor (pGP-AAV-syn-jGCaMP8m-WPRE) into the nTS. On the day of the experiment, an optic fiber was lowered on the surface of the brainstem above the nTS and calcium transient activity was recorded (Figure 6B). GCaMP viral expression was observed throughout the caudal-rostral extent of the nTS in all rats (Figure 6C). Fentanyl (20 µg/kg) infusion induced a biphasic activity profile in nTS neurons, characterized by an initial robust increase (Early Response) followed by a decrease below baseline (Late Response, Figure 6D). Pretreatment with NLXM (5 mg/kg) attenuated both phases of the fentanyl-induced changes in neuronal activity. As shown in Figure 6E, the area under the curve (AUC) was significantly elevated during the Early and Late Responses after fentanyl. In contrast, no significant fentanyl-induced change in AUC was observed in NLXM-pretreated rats at either time point, indicating that changes in fentanyl-induced nTS neuronal activity are strongly mediated by peripheral MORs. We next examined the extent to which MOR antagonists restore the fentanyl-induced reductions in activity. After MOR antagonist administration, fentanyl-induced suppression of nTS activity

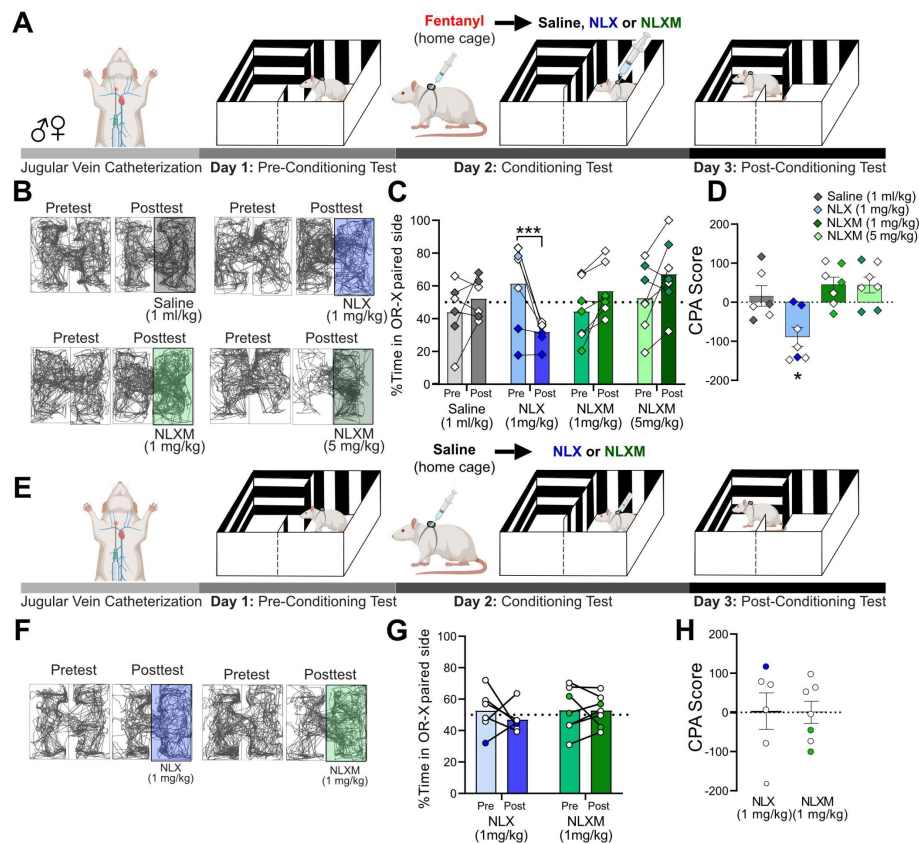


Figure 5

NLXM-mediated reversal of OIRD is not aversive.

A. Schematic of the Conditioned Place Aversion (CPA) experimental design for rats that received fentanyl followed by saline or an opioid receptor antagonist. Catheterized rats underwent a pre-conditioning test (Day 1). On the conditioning day (Day 2), a divider was placed in the middle of the CPA box. Rats received intravenous fentanyl (50 µg/kg) in their home cages. Four minutes later, rats were transferred to one side of the CPA box and immediately received intravenous saline (1 ml/kg, n=6), NLX (1 mg/kg, n=7), or NLXM (1 mg/kg, n=7; 5 mg/kg, n=7). During the post-conditioning test (Day 3), the divider was removed, and rats were placed in the box and allowed to explore both compartments. **B.** Track plot data showing movement in both sides of the CPA box during the Pretest (day 1) and Posttest (Day 3). For each condition, the Pre- and Posttest data is from the same animal. In these representative examples, the saline- or antagonist-paired side is shown on the right side of the box. However, the paired side was balanced between left and right sides of the box for all groups. **C.** Mean data with individual plots showing changes in the time spent in the saline- and antagonist-paired side. In all graphs, males are represented by filled symbols, and females by open symbols. There was a significant decrease in the time spent in the NLX-paired side: two-way repeated measures ANOVA; drug x test interaction: $F_{(3,23)} = 9.994$, $p = 0.0002$; Sidak's post-hoc, $***p = 0.0007$. **D.** The CPA score (calculated as time spent in the saline or antagonist paired compartment during posttest minus the time spent in the same compartment during the pretest). For CPA scores, one-sample t-test from hypothetical value (no aversion). There was a significant decrease in the CPA score for 1 mg/kg NLX treated rats ($t = 3.642$ (df=6), $*p < 0.05$). 1 mg/kg NLXM: $t = 2.340$, df=6 $p = 0.0578$; 5 mg/kg NLXM: $t = 2.172$, df=6, $p = 0.0729$. Saline, $t = 0.6113$ (df=5, $p = 0.5677$). **E.** Schematic of the Conditioned Place Aversion (CPA) experimental design for saline-pretreated animals. Catheterized rats underwent a pre-conditioning test (Day 1). On the conditioning day (Day 2), a divider was placed in the middle of the CPA box. Rats received intravenous saline (1 ml/kg) in their home cages. Four minutes later, rats were transferred to one side of the CPA box and immediately received intravenous NLX (1 mg/kg, n=6) or NLXM (1 mg/kg, n=7). **F.** Track plot data showing movement in both sides of the CPA box during the Pretest (day 1) and Posttest (Day 3). The paired side was balanced between left and right sides of the box for all groups. **G.** Mean data with individual plots showing changes in the time spent on the NLX- or NLXM-paired side. There was no difference in time spent in the side paired with NLX or NLXM (two-way repeated measures ANOVA, antagonist x time, $F_{(1,11)} = 0.4638$, $p = 5.099$). **H.** CPA scores for NLX- and NLXM-treated animals. There was no significant difference in the CPA scores in rats that received NLX (one sample t-test, $t = 0.07102$, df=5, $p = 0.9461$) or NLXM (one sample t-test, $t = 0.005032$, df=6, $p = 0.9961$).

returned to baseline during the early recovery in rats that received NLX, but not NLXM. However, both NLX and NLXM enhanced nTS activity above baseline during the late recovery period (**Figure 6F,G**). During the Late Recovery period, nTS activity was significantly enhanced above baseline in both NLX and NLXM-treated animals. Together, these data demonstrate that peripheral MOR antagonism is sufficient to reverse the fentanyl-induced decrease in nTS neuronal activity.

Discussion

OIRD is a complex problem that continues to be at the forefront of the opioid epidemic (1, 37). The increased potency and faster onset of synthetic opioids like fentanyl has been attributed to the rising wave of opioid-related deaths in the US (2, 11). Numerous studies have evaluated opioid-induced dysfunction within key regulatory sites in the central nervous system that contribute to the severity of OIRD (1, 15, 16, 18, 26). Peripheral MORs have also been examined for their roles in impacting OIRD (12, 34). However, the specific contributions of central and peripheral MORs to the onset and duration of fentanyl-induced cardiorespiratory depression, along with the role of the nTS in these effects, are not completely understood. In this study, we show that blocking peripheral MORs with NLXM sufficiently prevents and reverses fentanyl-induced cardiorespiratory depression and systemic hypoxia. Furthermore, we demonstrate that peripheral opioid receptors mediate fentanyl-induced activity of nTS neurons. These data suggest that the nTS may be an area impacted by fentanyl via peripheral MOR activation. Lastly, we demonstrate that in addition to reversing fentanyl-induced cardiorespiratory depression, NLXM does not induce aversive behaviors, as compared to NLX. Together, these findings provide evidence that peripheral MOR antagonism may be a potential novel strategy to reverse OIRD without inducing withdrawal, anxiety, and aversion, all of which can contribute to further relapse in drug-seeking behaviors (2, 13).

While numerous studies have focused on evaluating the effects of opioids acting at MORs present within brainstem and pontine networks, less attention has been given to identifying peripheral MOR mechanisms contributing to OIRD. NLX is the most common reversal agent used to treat OIRD (13, 37). In addition to reversing opioid-induced analgesia, NLX precipitates withdrawal symptoms and aversion in humans and rodents (12, 13, 38, 39). In this study, we demonstrate that NLX-mediated reversal of OIRD caused a conditioned place aversion, as shown by a significant decrease in time spent in the NLX-associated compartment. This is likely the result of the induction of withdrawal symptoms that occur following antagonism of MOR in central reward pathways (12, 13, 40). In contrast, this aversive response was not observed in rats that received NLXM. These data suggest that alleviating fentanyl-induced respiratory distress without antagonizing central MOR may mitigate the aversive state produced by NLX.

Conditioned place preference and aversion paradigms have been utilized to evaluate rewarding and aversive properties of opioids and antagonists (41–43). Overall, we observed relatively minimal sex differences in basal and fentanyl-induced changes in cardiorespiratory depression, which is consistent with a previous report showing that sex differences due to fentanyl are relatively minimal compared to other opioids such as heroin (9). Previous reports have shown sexually dimorphic differences in fentanyl-induced conditioned place preference, with higher doses of fentanyl affecting females only (43). In present study, we administered fentanyl to male and female rats in the home cage followed by saline in the CPA box and observed neither a preference nor aversion in either sex. This suggests that the results obtained during OIRD-mediated reversal via NLX or NLXM are driven by effects within the CPA box and are not influenced by the rewarding properties of fentanyl itself. While previous reports have shown that NLX can be detected centrally after a subcutaneous injection of NLXM (33), we observed no detectable amount of NLX or NLXM in brain tissue of rats that received intravenous NLXM at either dose or time point. Our data demonstrate robust fentanyl-induced neuronal activation in the nTS that appears to be mediated via peripheral MORs. Given the proximity of the nTS to the

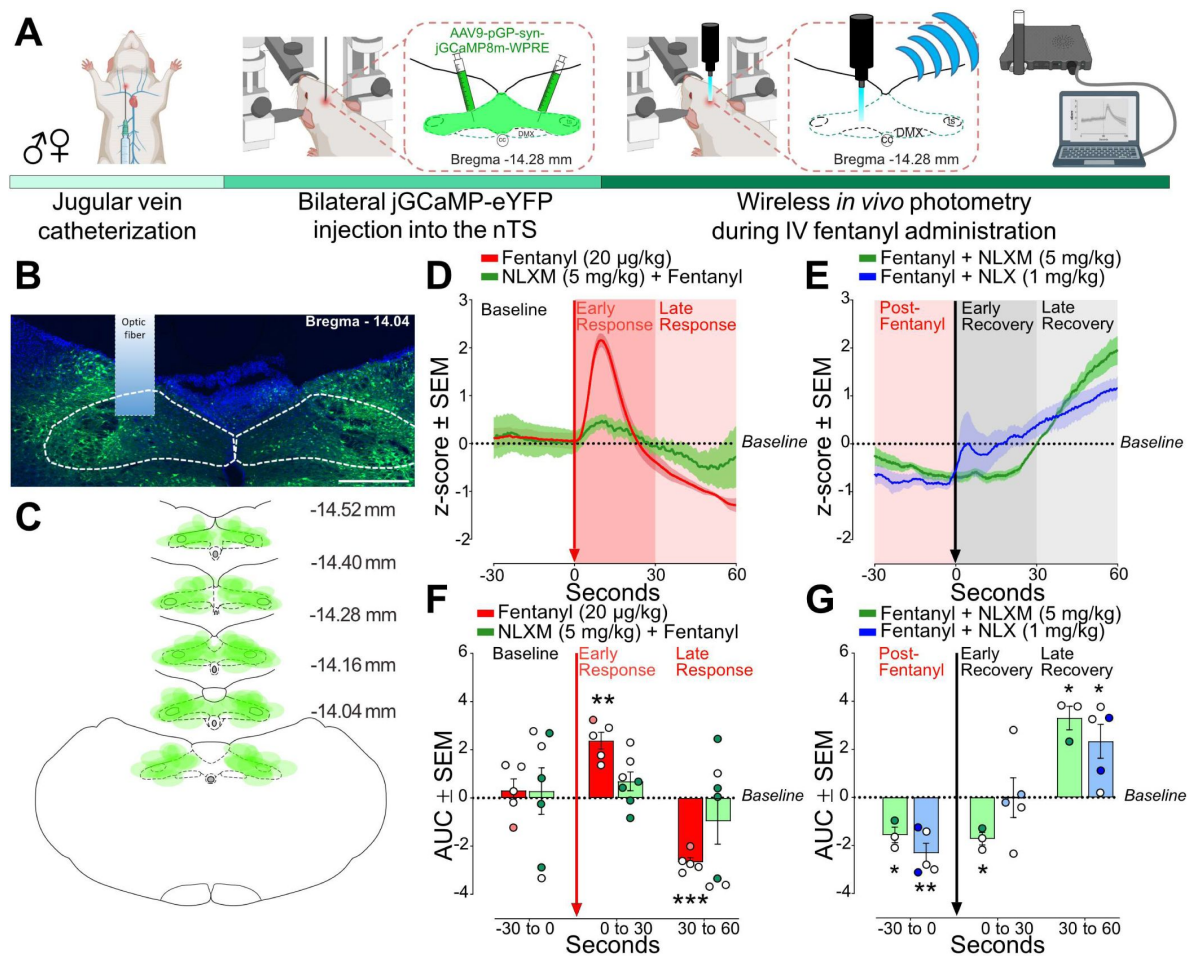


Figure 6

Fentanyl-induced nTS activity is mediated by peripheral MORs.

A. Experimental timeline. Male and female rats underwent jugular catheterization surgery, followed by bilateral nanoinjection of jGCaMP8m into the nTS and then allowed 3-5 weeks for viral expression. On the test day, rats were anesthetized and placed in a stereotaxic apparatus. The nTS was exposed and a fiber was lowered into the nTS. **B.** Representative example of GCaMP expression and fiber placement in the nTS. Scale bar = 200 μ m. **C.** Spread of overlay of individual animal viral expression across the caudal-rostral extent of the nTS. **D.** nTS cell response (mean z-score) aligned to 20 μ g/kg fentanyl administration (red arrow) at Baseline (-30 to 0 seconds), early response (0 to 30 seconds) and late response (30 to 60 seconds) in rats that received 20 μ g/kg fentanyl only (red trace) or 5 mg/kg NLXM followed by 20 μ g/kg fentanyl (green trace). In all graphs, males are represented by filled symbols, and females by open symbols. **E.** Area under the curve (AUC) was calculated during baseline (-30 to 0 seconds), Early Response (0 to 30 seconds) or Late Response (30 to 60 seconds). For AUC graphs, one-sample t-test from hypothetical zero value. In rats that received fentanyl only, there was a significant increase in the AUC during the Early Phase ($t=6.747$, $df=4$, $**p=0.0025$) followed by a significant decrease in the Late Phase ($t=14.35$, $df=4$, $***p=0.0001$). In rats that received 5 mg/kg NLXM prior to 20 μ g/kg fentanyl, no differences in AUC were observed at any time point (Baseline; $t=0.2916$, $df=6$ $p=0.7804$; Early Phase; $t=1.760$, $df=6$ $p=0.1288$; 30-60 seconds; $t=1.002$, $df=6$, $p=0.3548$). **F.** nTS cell response (mean z-score) aligned to MOR antagonists (black arrow). Rats received iv 20 μ g/kg fentanyl (Post-Fentanyl phase, -30 to 0 seconds) prior to MOR antagonist infusion of NLX (1 mg/kg, blue trace) or NLXM (5 mg/kg, green trace). Data during Early Recovery (0 to 30 seconds) and Late Recovery (30 to 60 seconds) are shown. **G.** Mean data showing AUC analyses for both groups. For all graphs, one-sample t-test from hypothetical zero value. NLXM-treated rats had a significant decrease in activity from Post-Fentanyl baseline prior to NLXM ($t=4.799$, $df=2$ $*p=0.0408$) and during the Early Recovery phase ($t=6.683$, $df=2$ $p=0.0217$), and exhibited enhanced activity above baseline during the Late Recovery phase ($t=6.716$, $df=2$, $*p=0.0215$). NLX-treated rats displayed a similar decrease below baseline prior to NLX. No difference from baseline was observed during the Early Recovery phase ($t=0.01238$, $df=4$ $p=0.9907$), and a significant increase above baseline was observed during the Late Recovery phase ($t=3.318$, $df=4$, $*p=0.0294$).

circumventricular organ area postrema (44), this raises the possibility that fentanyl can access the central nervous system via the area postrema to influence neuronal activation. However, our data showing the lack of a conditioned place aversion following NLXM-mediated reversal of OIRD support our biodistribution data showing no NLXM detection in brain samples that the observed effects primarily involve fentanyl actions at MOR located in peripheral sites. Similar to NLX, peripheral MOR antagonism with NLXM has been shown to partially reverse opioid-induced analgesia (12, 45) which may pose limitations for pain management. While we report no changes in cardiorespiratory parameters from baseline following intravenous administration of NLX or NLXM in drug-naïve animals, previous work has shown that high doses of NLX elicit behavioral and physiological changes in humans, including cognitive impairment, nausea, tachycardia and hypertension, even at low and moderate doses (14, 46, 47). Future investigations are needed to evaluate mechanisms underlying the therapeutic potential of NLXM as an intervention for managing OIRD in patients with OUDs.

The nTS is a region with a high degree of MOR expression, and appears to be primarily located on afferent fibers arising from the vagus nerve (7, 25, 48). While MOR has been shown to be present in nTS somas and dendrites (25, 26), these MOR-expressing cells do not express Fos following exposure to opioids or hypoxia (26), which suggests a mechanism of presynaptic action of MOR signaling in the nTS. Likewise, the nodose ganglion (NG), which contains the cell bodies of nTS-projecting vagal sensory afferents, has high expression of MOR (32), and intra-nodose injection of fentanyl induces hypoventilation, although to a much lesser degree compared to systemic fentanyl (49).

In the present study, we used fiber photometry to characterize nTS neuronal responses to show that fentanyl induces an initial excitatory response in nTS cells. However, this response was short-lasting compared to the subsequent decrease in activity below baseline, suggesting that the primary effect of MOR signaling produce generalized inhibition of nTS neurons. This prolonged suppression of nTS neuronal activity may contribute to the overall duration of OIRD. Furthermore, our photometry data demonstrate that while the NLXM-mediated reversal of fentanyl-induced inhibition of nTS neurons was slower compared to NLX, activity was eventually restored. A similar finding was observed in OIRD experiments, as NLXM induced a rate of recovery in all parameters that was comparable to NLX. Taken together, this suggests that peripheral MOR antagonism may sufficiently override simultaneous central effects of fentanyl acting at MOR present in cardiorespiratory nuclei, including the nTS.

It is important to note that while all our experiments assessing fentanyl-induced cardiorespiratory depression were conducted in conscious rats breathing room air, photometry experiments were performed under isoflurane anesthesia. Due to the location of the nTS, chronic fiber implantation and subsequent in vivo photometry recordings in conscious, unrestrained animals are not possible. In the nTS, Isoflurane has been reported to suppress visceral afferent glutamatergic transmission and enhance GABAergic transmission via postsynaptic mechanisms (50), as well as induce cFos expression in the nTS (51). Precautions were taken to use the lowest dose of Isoflurane needed to maintain sufficient anesthesia.

However, we cannot rule out the possible influence of Isoflurane anesthesia to influence the observed biphasic responses evoked by fentanyl. In addition, sex differences in physiological and pharmacological responses to anesthesia have been previously described (52). Due to the high attrition rate for these experiments, which require multiple survival surgeries and proper fiber placement into the nTS, we were unable to fully evaluate potential sex differences and cannot rule out the possibility of a sex effect of fentanyl-induced neural activity or in responses to anesthesia. Nevertheless, our data support a mechanism by which fentanyl induces OIRD via disruption of visceral afferent signaling to the nTS, thereby leading to prolonged OIRD due to suppression of physiological reflex responses that would be normally engaged in response to systemic hypoxia.

The nTS plays an essential role in the regulation of basal-and reflex-evoked cardiorespiratory function (22). Activation of MOR in the nTS has been shown to blunt hypoxic ventilatory responses following MOR agonism in the nTS (7). Our data are consistent with previous work demonstrating opioid-induced activation of nTS neurons (26). The degree of fentanyl induced Fos-IR in the nTS resembles what is observed after acute exposure to hypoxic air (20, 46). Interestingly, we also observed increased Fos in the nTS at a dose of fentanyl that was below the threshold to induce respiratory depression. It is unclear whether the Fos-IR observed after low or high doses of fentanyl is, in part, the result of increased glutamatergic transmission stemming from enhanced chemoreceptor afferent signaling to the nTS or via MOR signaling within the nTS itself.

Glutamate is the primary neurotransmitter released from sensory afferent fibers into the nTS (22, 53). While intra-nTS application of opioid agonists have been shown to reduce glutamate release and calcium channels via presynaptic mechanisms (54, 55), opioids also have been shown to induce a similar biphasic action of Ca²⁺-dependent spikes in the nodose ganglion (56) that resembles what we observed in GCaMP-expressing nTS cells after intravenous fentanyl. NLXM strongly attenuated the fentanyl-induced excitatory wave observed in GCaMP-expressing nTS neurons, suggesting that MOR-expressing peripheral afferents mediate this transient activation of nTS neurons, even if the primary effect of fentanyl is to inhibit the nTS and suppress cardiorespiratory reflex function.

One possibility is that the increase in nTS activity may be mediated by opioid-induced disinhibition within the nTS. The nTS contains a large population of second order GABAergic interneurons that receive direct inputs from visceral afferents (57), and hypoxia activates these cells (23). nTS MOR agonists suppress the activity of these cells, primarily via presynaptic mechanisms (58), although postsynaptic mechanisms have been reported (59). Similarly, visceral afferents also activate catecholaminergic nTS neurons (60), which are critical for the generation of hypoxia-evoked cardiorespiratory responses (35, 61). Opioids inhibit visceral afferent transmission to this cell group, primarily via a presynaptic mechanism (54). In the present study, we observed robust Fos expression in these cells at all doses of fentanyl examined. These findings raise the possibility that nTS dysfunction, via MOR-expressing peripheral inputs that terminate within this region, represents the first central site of fentanyl-induced dysfunction, ultimately leading to downstream effects in other brain regions that contribute to diminished autonomic and cardiorespiratory responses that contribute to the duration of OIRD. Future studies are needed to fully evaluate the exact opioid-MOR effects in the nTS, including specific MOR afferent inputs and the nTS phenotypes, to determine their relative contributions to OIRD.

In summary, these studies highlight a significant involvement of peripheral MORs contribute to the rapid cardiorespiratory depression induced by intravenous fentanyl. We have provided extensive characterization into how fentanyl elicits cardiorespiratory depression and examined the relationship between peripheral MORs and nTS activity in contributing to the duration of OIRD. Moreover, our findings suggest that selectively blocking peripheral opioid receptors could be a promising therapeutic strategy for managing OIRD. Notably, peripheral opioid receptor antagonists are already prescribed for patients undergoing chemotherapy to alleviate side effects such as opioid-induced constipation (62, 63). Importantly, our data demonstrate that peripheral MOR antagonism appears to reverse OIRD without triggering unwanted effects of withdrawal that occur following NLX administration, suggesting that peripheral MOR antagonists like NLXM could potentially be utilized in treatment strategies for managing patients with OUD.

Materials and Methods

All surgical and experimental procedures were approved by Washington University Committee in accordance with the National Institutes of Health Guidelines for the Care and Use of Laboratory Animals and Animal Research: Reporting In Vivo Experiments (ARRIVE) guidelines. Rats were initially group-housed with two to three animals per cage on a 12/12-h dark/light cycle (lights on at 7:00) and acclimated to the animal facility holding rooms for at least 7 days before any manipulation. Following catheterization surgery, rats were single-housed and remained on an identical 12/12-h dark/light cycle. The temperature for the holding rooms of all animals ranged from 21 to 24 °C while the humidity was between 30 and 70%. Rats received food and water ad libitum for the duration of all experiments. All experiments were performed during the light cycle. For all experiments, male and female Sprague Dawley rats (250-350g) were used. In all data sets, male rats are represented by filled symbols and female rats are represented by open symbols.

IV catheterization procedure

Animals were deeply anesthetized using Isoflurane (3% induction, 2% maintenance). A small incision was made in the dorsal surface of the neck. Another small incision was made on the ventral surface of the neck, and underlying tissue was carefully dissected to expose the jugular vein. An indwelling catheter was inserted into the jugular vein and sutured in place using non-absorbable silk suture. The catheter was then tunneled subcutaneously and exited the body through the first small hole in neck on the dorsal side. The ventral incision was closed using non-absorbable silk suture. The exposed catheter was connected to a backpack device (Instech) containing a port for drug administration. Animals were given Carprofen (2 mg/kg sc), Baytril (8 mg/kg sc) and bupivacaine (5 mg/kg, site of incision). In addition, rats were given Carprofen tablets (Bio Serv, MD150-2) for 2 days after surgery to assist in wound healing and analgesia. Rats were singly housed following surgery and were allowed to recover for one week prior to the commencement of experiments. Catheter patency was maintained with daily flushing of 0.3 mL sterile saline containing gentamicin (1.33 mg/mL, i.v.)

Respiratory depression experiments

Following recovery from IV catheterization surgery, conscious, freely moving rats in their home cages were fitted with a non-invasive pulse oximeter collar (Starr LifeSciences). Cardiorespiratory parameters (oxygen saturation, heart rate and respiratory rate) were collected (MouseOx v2.0, Starr LifeSciences) at baseline conditions in animals breathing room air, and after intravenous administration of fentanyl citrate (2, 20 or 50 µg/kg). A subset of animals received weight-adjusted intravenous administration of the nonselective opioid receptor antagonist naloxone hydrochloride (NLX; 1 mg/kg; Sigma 51481-60-8) or the peripherally restricted opioid receptor antagonist naloxone methiodide (NLXM; 1 or 5 mg/kg; Sigma N129) before or after fentanyl administration. Cardiorespiratory parameters were then measured for up to 60 minutes. For all OIRD experiments, the time course graphs consist of 30 seconds of data averaged into a single data point. The nadir was defined as the lowest point measured after fentanyl administration (for all graphs, red arrow indicates time of fentanyl administration, time=0 minutes). In addition, the recovery rate was defined as the time to reach 90% baseline of pre-fentanyl values for each cardiorespiratory parameter. Two hours after fentanyl administration, rats were transcardially perfused with 0.01M PBS followed by paraformaldehyde (4% in 0.01M PBS). Brains were removed and stored overnight at 4°C in 4% paraformaldehyde for post-fixation, followed by at least 72 h incubation in sucrose (30% in 0.01M PBS) solution. Isopentane was used to flash-freeze brains. Brainstem sections containing the nTS were sliced in the coronal plane (40 µm) using a cryostat (Leica CM 1950).

Biodistribution

A subset of previously catheterized drug-naïve rats was deeply anesthetized and then received intravenous administration of NLX (1 mg/kg) or NLXM (1 or 5 mg/kg). While still under anesthesia, rats were quickly decapitated two or ten minutes after intravenous administration of MOR antagonists. MOR antagonists were dosed in all rats at an experimentally relevant concentration with triplicate blood and brain samples collected at two time points. The early time point at 2 minutes evaluated conditions when the plasma concentration of the OR antagonist is high and reflects the identical time point of intravenous administration for OIRD pretreatment experiments. The second collection time point (10 minutes) was intended to evaluate steady-state conditions and represent physiological experiments when reversal of OIRD was achieved by both antagonists. Brains were quickly removed and flash frozen in liquid nitrogen. Trunk blood was collected into capillary collection tubes containing lithium heparin (Greiner Bio-One; Fisher Scientific). The blood was spun on a centrifuge for 10 min at 4 °C, and plasma was carefully extracted from the supernatant. Brains and plasma samples underwent PK analyses for detection of NLX or NLXM at both time points. Standard rat PK studies dosed the candidate MOR antagonists and sample blood over multiple time points. A general description of tissue distribution can be obtained during PK modeling by calculating the volume of distribution. Samples underwent a direct measurement of peripheral restriction by measuring compound concentration in plasma and brain.

Conditioned Place Aversion

The aversive effects following reversal of OIRD were measured using Conditioned Place Aversion (CPA) (64 [DOI](#)). During a pre-conditioning session (Day 1), rats were allowed to explore both sides of a two compartment CPA box for 15 minutes. On the Conditioning Day (Day 2), a divider was placed in the middle of the box to restrict movement to one side on the box. Rats were then randomly assigned to one side of the CPA box, and this side was paired with either saline (1 ml/kg), NLX (1 mg/kg) or NLXM (1 or 5 mg/kg). Rats were given intravenous fentanyl (50 µg/kg) or saline in their home cages. Four minutes later, they were placed in the paired side of the box and immediately given saline or a MOR antagonist. Rats were confined to this compartment for 15 minutes. For all CPA experiments, measures were taken to balance the side of CPA box by drug and by sex. On the post-conditioning test (Day 3), the divider was removed. Rats were then placed in the CPA box and allowed to explore both sides of the CPA box for 15 minutes. The total time spent in the paired chamber and CPA scores were calculated for all groups.

Wireless *in vivo* Fiber Photometry

To selectively measure nTS neuronal calcium transient activity, we injected an AAV containing a calcium sensor expressed under the synapsin promoter (pGP-AAV-syn-jGCaMP8m-WPRE) into the nTS (500 nl per side; 0.3-0.5 mm anterior, ± 0.4 mm lateral and 0.4 mm ventral to brain surface, relative to calamus scriptorius). Rats were allowed at least 3 weeks to recover from nanoinjection surgery to allow for robust viral expression in the nTS before IV catheters were implanted as described above. On the day of the experiment, rats were deeply anesthetized with isoflurane (3-4% induction, 2% maintenance), placed in the stereotaxic apparatus, and the nTS was exposed as described above. To overcome limitations associated with chronic fiber implants in the nTS, a wireless photometry headstage (TeleFipho, Amuza Inc.) was secured to a 5 mm flat-tipped silica optic fiber cannula (400 µm c.d./470 µm o.d., NA 0.37, 2.5 mm receptacle; Doric) and mounted the stereotaxic arm with an apparatus 3D printed in house to permit simultaneous nTS exposure and photometry recordings. The optic fiber was lowered to the site of the dorsal brainstem and firmly pressed on the surface above the nTS until a signal was detected (~0.3-0.5 mm ventral from the surface). An LED generated blue light was bandpass-filtered (445-490 nm) to excite jGCaMP8m and emission fluorescence was detected by an internal photodiode detector for green light (bandpass filtered at 500-550 nm). An internal DC amplifier transmitted the data (16-bit arbitrary unit; AU) to a wireless receiver (TeleFipho, Amuza Inc.) and the data was extracted in real-time using

TeleFipho software (Amuza Inc.) at a sampling rate of 100 Hz. The offset was set to 90° and the LED power was adjusted to achieve an optimal signal output (30–40k a.u.) not to exceed 36 mW power, and remained constant throughout the recording. Baseline activity of GCaMP fluorescence nTS-infected cells was measured after at least 5 min to mitigate effects of baseline drift in signal due to slow photobleaching artifacts. Then, baseline activity was measured for at least 30 s prior to administration of either fentanyl (20 µg/kg, i.v.) or NLXM (5 mg/kg, i.v.). After response activity was recorded, rats received a second intravenous injection of either an opioid receptor antagonist (NLX, 1 mg/kg or NLXM, 5 mg/kg) or fentanyl (20 µg/kg), respectively. Each intravenous administration was manually time locked to calcium transient recordings by a second experimenter and response activity following each injection was recorded for 60 s before administration of the second injection or conclusion of the experiment. Rats were transcardially perfused with PFA at the end of the experiment as described above and viral expression was validated for all animals.

For analysis of calcium transient activity, the raw data was fit to a double-exponential curve and subtracted from the raw data to account for any residual effects of baseline drift. Matlab Signal Analyzer was used to extract regions of interest (ROIs) in the signal in 90-s traces (30 s pre-treatment, 60 s post-treatment) and traces were aligned to the time of injection. ROIs were standardized as z-scores using the sample standard deviation, S , where $S = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$ such that signals were centered to have a mean of 0 and scaled to have a standard deviation of 1. Based on this, the area under the curve (AUC) of ROI z-scores were calculated in 30-s bins using the trapezoidal rule (*trapz* function in Matlab) and one-sample t-tests were used to determine differences relative to baseline (theoretical mean = 0). For presentation, transients were down sampled by a factor of 10 with a phase offset of 2 and the resultant 10 Hz traces were smoothed using a moving mean duration of 5% of total ROI.

Immunohistochemistry (IHC)

Immunohistochemical procedures were performed as previously described (65). Brainstem sections containing the nTS were sliced using a cryostat (Leica) at a thickness of 40 µm. Free-floating sections were rinsed in 0.01M PBS, blocked in 10% Normal Donkey Serum (Sigma) in 0.01M PBS-triton, and then incubated with the following antibodies: cFos (Rabbit anti-cFos; ab190289, Abcam; AB_2737414); Tyrosine Hydroxylase (TH, Mouse anti-TH; MAB318, Millipore; AB_2201528). The following day, sections were rinsed with 0.01M PBS, incubated with Donkey anti-Rabbit Cy3 and Donkey anti-Mouse Alexa fluor 488 (1:200; in 3% NDS and 0.3% Triton in 0.01M PBS) and used to visualize targets of interest (Jackson ImmunoResearch).

Image acquisition

Immunoreactivity (IR) and viral expression were examined with a Leica DMR microscope at 20X magnification. 20 µm thick z-stacks of the nTS (2 µm per plane) were collected. ImageJ (v. 1.48, NIH) software was used for post-processing and analysis. In all groups, three sections of the nTS (corresponding to −360, 0, +360 µm relative to calamus scriptorius) were collected.

Brainstem immunohistochemical analyses

Using ImageJ software, the regions containing the nTS were outlined, and unilateral counts of Fos-IR, TH-IR and co-labeled Fos- and TH-IR neurons were performed using a custom-made plugin (Cell Counter) for each of the 3 nTS sections per animal. The total number of each phenotype was summed from these 3 sections, and the percentage of TH-IR cells co-expressing Fos-IR was determined.

Statistics

For all experiments, the normality of sample data was determined using D'Agostino and Pearson tests and Shapiro–Wilk tests. Statistical significance was taken as * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ and **** $p < 0.0001$, as determined by two-way repeated measures analysis of variance (ANOVA) followed by Sidak's post hoc test; one-way ANOVA with Tukey post-hoc test, two-tailed unpaired t-test, two-tailed paired t-test, one sample t-test.

Data Availability Statement

The authors confirm that all relevant data are included in the paper and its Supplementary Information files. All data generated in this study are provided in the Supplementary Information/Source Data file. Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Jose A. Morón (jmoron-concepcion@wustl.edu).

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Conceptualization: BCR NM and JAM

Methodology: BCR, NM and JAM

Formal analysis of all data: BCR, SM, RK, MT and JAH

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Respiratory depression experiments: BCR, SM, RK, MT, JM, CGR, SAL

Conditioned Place Aversion: BCR, SM, MT, JM

Biodistribution experiments: BCR and TL

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Additional files

Supplemental Figures S1-S3 and Tables 1-4 [↗](#)

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Reviewer #1 (Public review):

Summary:

This paper shows that the synthetic opioid fentanyl induces respiratory depression in rodents. This effect is reversed by the opioid receptor antagonist naloxone, as expected. Unexpectedly, the peripherally restricted opioid receptor antagonist naloxone methiodide also blocks fentanyl-induced respiratory depression.

Strengths:

The paper reports compelling physiology data supporting the induction of respiratory distress in fentanyl-treated animals. Evidence suggesting that naloxone methiodide reverses this respiratory depression is compelling. This is further supported by pharmacokinetic data suggesting that naloxone methiodide does not penetrate into the brain, nor is it metabolized into brain-penetrant naloxone.

Weaknesses:

The paper would be further strengthened by establishing the functional significance of the altered neural activity detected in the nTS (as measured by cFos and GcAMP/photometry) in the context of opioid-induced respiratory depression.

<https://doi.org/10.7554/eLife.104469.2.sa3>

Reviewer #2 (Public review):

Summary:

In this article, Ruyle and colleagues assessed the contribution of central and peripheral mu opioid receptors in mediating fentanyl-induced respiratory depression using both naloxone and naloxone methiodide, which does not cross the blood brain barrier. Both compounds prevented and reversed fentanyl-induced respiratory depression to a comparable degree. The advantage of peripheral treatments is that they circumvent the withdrawal-like effects of naloxone. Moreover, neurons located in the nucleus of the solitary tract are no longer activated by fentanyl when naloxone methiodide is administered, suggesting that these responses are mediated by peripheral mu opioid receptors. The results delineate a role for peripheral mu opioid receptors in fentanyl-derived respiratory depression and identify a potentially advantageous approach to treating overdoses without inflicting withdrawal on the patients.

Strengths:

The strengths of the article include the intravenous delivery of all compounds, which increases the translational value of the article. The authors address both prevention and

reversal of fentanyl-derived respiratory depression. The experimental design and data interpretation are rigorous and appropriate controls were used in the study. Multiple doses were screened in the study and the approaches were multipronged. The authors demonstrated activation of NTS cells using multiple techniques and the study links peripheral activation of mu opioid receptors to central activation of NTS cells. Both males and females were used in the experiments. The authors demonstrate the peripheral restriction of naloxone methiodide.

Weaknesses:

Naloxone is already broadly used to prevent overdoses from opioids so in some respects, the effects reported here are somewhat incremental.

Comments on the latest version:

I think the authors have adequately addressed previous critiques and I don't have any additional comments.

<https://doi.org/10.7554/eLife.104469.2.sa2>

Reviewer #3 (Public review):

Summary

This manuscript outlines a series of very exciting and game-changing experiments examining the role of peripheral MORs in OIRD. The authors outline experiments that demonstrate a peripherally restricted MOR antagonist (NLX Methiodide) can rescue fentanyl-induced respiratory depression and this effect coincides with a lack of conditioned place aversion. This approach would be a massive boon to the OUD community, as there are a multitude of clinical reports showing that naloxone rescue post fentanyl over-intoxication is more aversive than the potential loss-of-life to the individuals involved. This important study reframes our understanding of successful overdose rescue with a potential for reduced aversive withdrawal effects.

Strengths:

Strengths include the plethora of approaches arriving at the same general conclusion, the inclusion of both sexes, and the result that a peripheral approach for OIRD rescue may side-step severe negative withdrawal symptoms of traditional NLX rescue.

Weaknesses:

All weaknesses were addressed.

<https://doi.org/10.7554/eLife.104469.2.sa1>

Author response:

The following is the authors' response to the original reviews.

eLife Assessment

This manuscript represents a fundamental contribution demonstrating that fentanyl-induced respiratory depression can be reversed with a peripherally-restricted mu opioid receptor antagonist. The paper reports compelling and rigorous physiological, pharmacokinetic, and behavioral evidence supporting this major claim, and furthers

mechanistic understanding of how peripheral opioid receptors contribute to respiratory depression. These findings reshape our understanding of opioid-related effects on respiration and have significant therapeutic implications given that medications currently used to reverse opioid overdose (such as naloxone) produce severe aversive and withdrawal effects via actions within the central nervous system.

We thank the reviewers for their insightful comments and critiques, which we have incorporated into the manuscript. We believe these revisions have significantly improved the manuscript. Additionally, following discussions among the authors, we have revised the color scheme across all figures. For example, the color of the symbols in Figure 1B-D now match the bars in Figure 1E-J, rather than the symbols. We feel that this change improves the clarity and visual consistency of the figures, making it easier to interpret the data across figures.

Public Reviews:

Reviewer #1 (Public review):

Summary:

This paper shows that the synthetic opioid fentanyl induces respiratory depression in rodents. This effect is reversed by the opioid receptor antagonist naloxone, as expected. Unexpectedly, the peripherally restricted opioid receptor antagonist naloxone methiodide also blocks fentanyl-induced respiratory depression.

Strengths:

The paper reports compelling physiology data supporting the induction of respiratory distress in fentanyl-treated animals. Evidence suggesting that naloxone methiodide reverses this respiratory depression is compelling. This is further supported by pharmacokinetic data suggesting that naloxone methiodide does not penetrate into the brain, nor is it metabolized into brain-penetrant naloxone.

Weaknesses:

A weakness of the study is the fact that the functional significance of opioid-induced changes in neural activity in the nTS (as measured by cFos and GcAMP/photometry) is not established. Does the nTS regulate fentanyl-induced respiratory depression, and are changes in nTS activity induced by naloxone and naloxone methiodide relevant to their ability to reverse respiratory depression?

Reviewer #2 (Public review):

Summary:

In this article, Ruyle and colleagues assessed the contribution of central and peripheral mu opioid receptors in mediating fentanyl-induced respiratory depression using both naloxone and naloxone methiodide, which does not cross the blood-brain barrier. Both compounds prevented and reversed fentanyl-induced respiratory depression to a comparable degree. The advantage of peripheral treatments is that they circumvent the withdrawal-like effects of naloxone. Moreover, neurons located in the nucleus of the solitary tract are no longer activated by fentanyl when naloxone methiodide is administered, suggesting that these responses are mediated by peripheral mu opioid receptors. The results delineate a role for peripheral mu opioid receptors in fentanyl-derived respiratory depression and identify a potentially advantageous approach to treating overdoses without inflicting withdrawal on the patients.

Strengths:

The strengths of the article include the intravenous delivery of all compounds, which increase the translational value of the article. The authors address both the prevention and reversal of fentanyl-derived respiratory depression. The experimental design and data interpretation are rigorous and appropriate controls were used in the study. Multiple doses were screened in the study and the approaches were multipronged. The authors demonstrated the activation of NTS cells using multiple techniques and the study links peripheral activation of mu opioid receptors to central activation of NTS cells. Both males and females were used in the experiments. The authors demonstrate the peripheral restriction of naloxone methiodide.

Weaknesses:

Naloxone is already broadly used to prevent overdoses from opioids so in some respects, the effects reported here are somewhat incremental.

The reviewer is correct that naloxone is the standard antidote for reversing opioid-induced respiratory depression. However, its limitations, including the risk of precipitated withdrawal, are well-documented in both preclinical and clinical studies. The likelihood of withdrawal increases when multiple doses of naloxone are administered. Since naloxone-induced withdrawal is centrally mediated, this study aimed to evaluate a peripherally restricted MOR antagonist for its ability to prevent or reverse fentanyl-induced respiratory depression. A key finding is that NLXM reversed OIRD without inducing aversive behavior. This suggests that peripheral antagonists like NLXM may be integrated into intervention strategies that save lives while preventing the adverse behavioral and physiological effects that are observed after treatment with naloxone.

Reviewer #3 (Public review):

Summary:

This manuscript outlines a series of very exciting and game-changing experiments examining the role of peripheral MORs in OIRD. The authors outline experiments that demonstrate a peripherally restricted MOR antagonist (NLX Methiodide) can rescue fentanyl-induced respiratory depression and this effect coincides with a lack of conditioned place aversion. This approach would be a massive boon to the OUD community, as there are a multitude of clinical reports showing that naloxone rescue post fentanyl over-intoxication is more aversive than the potential loss-of-life to the individuals involved. This important study reframes our understanding of successful overdose rescue with potential for reduced aversive withdrawal effects.

Strengths:

Strengths include the plethora of approaches arriving at the same general conclusion, the inclusion of both sexes and the result that a peripheral approach for OIRD rescue may side-step severe negative withdrawal symptoms of traditional NLX rescue.

Weaknesses:

The major weakness of this version relates to the data analysis assessed sex-specific contributors to the results.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Some points for the authors to consider are:

(1) In the Abstract, it is unclear why "high potency and lipophilicity" contribute to opioid-induced respiratory depression.

The higher potency of fentanyl compared to other opioids significantly increases the risk of overdose and subsequent respiratory depression. Its high lipophilicity facilitates rapid absorption and central nervous system penetration, which contributes to the rapid onset of these cardiorespiratory depression. The narrow therapeutic window of fentanyl further emphasizes the critical need for timely intervention when an overdose has occurred, and effective antagonists to reverse respiratory depression and save lives. We have revised the abstract to clarify these points.

(2) Are the doses of fentanyl used in the study (2, 20, or 50 µg/kg IV) relevant to those achieved by fentanyl-exposed human drug users?

In these studies, we intravenously administered three doses of fentanyl. The human equivalent doses (HED) of 20 µg/kg and 50 µg/kg fentanyl are ~3 µg/kg and ~8 µg/kg, respectively. These doses have previously been shown to induce respiratory depression in humans (Dahan et al., 2005).

(3) In Figure 1, it appeared that only a small fraction of tyrosine hydroxylase-positive (TH+) neurons expressed cFos in response to fentanyl, and the degree of cFos expression was largely similar across all fentanyl doses tested. Thus, it is unclear whether TH+ neurons play a role in fentanyl-induced respiratory depression, and the value of these data is unclear (see point #6 below also).

As shown in the mean data, the lowest dose of fentanyl, which was below the threshold for inducing OIRD, activated approximately 50% of tyrosine hydroxylase-positive (TH+) nTS neurons. In contrast, the highest dose of fentanyl resulted in a statistically significant increase, with ~75% of TH+ cells co-expressing Fos-IR.

We included the assessment of catecholaminergic nTS cells for several reasons. The regions of the nTS evaluated in this study contains high expression of MOR and are the termination points of sensory afferent fibers transmitting cardiorespiratory information to the nTS (Aicher et al., 2000; Furdul et al., 2024). Catecholaminergic cells receive direct excitatory inputs from visceral afferents (Appleyard et al., 2007) and exhibit intensity-dependent increases in Fos-IR in rats exposed to hypoxic air (Kline et al., 2010; King et al., 2012). These neurons are essential for generating appropriate cardiorespiratory responses to hypoxic challenges (Bathina et al., 2013; King et al., 2015). As the reviewer notes, rats exposed to fentanyl exhibit a high degree of Fos-IR in the nTS, including catecholaminergic neurons. Despite the robust fentanyl-induced activation (increased Fos-IR) nTS neurons, yet there appears to be a failure to initiate appropriate chemoreflex-mediated cardiorespiratory responses. Our photometry data further indicate that fentanyl-induced changes in neuronal activity are mediated, in part, by peripheral MOR. Collectively, these findings suggest that fentanyl impacts nTS activity through alterations in peripheral afferent signaling to the nTS, which may contribute to the severity and duration of OIRD.

(4) It would help with the flow of the paper if the pharmacokinetic data shown in Figure 6 were presented earlier (as part of Figure 2).

We have moved the biodistribution data earlier in the manuscript, now presenting it as Figure 2. The numbering of all subsequent figures has been adjusted accordingly.

(5) In Figure 5, there appears to be a large number of GCaMP-expressing neurons located outside the nTS. To what degree can the changes in calcium signaling, attributed

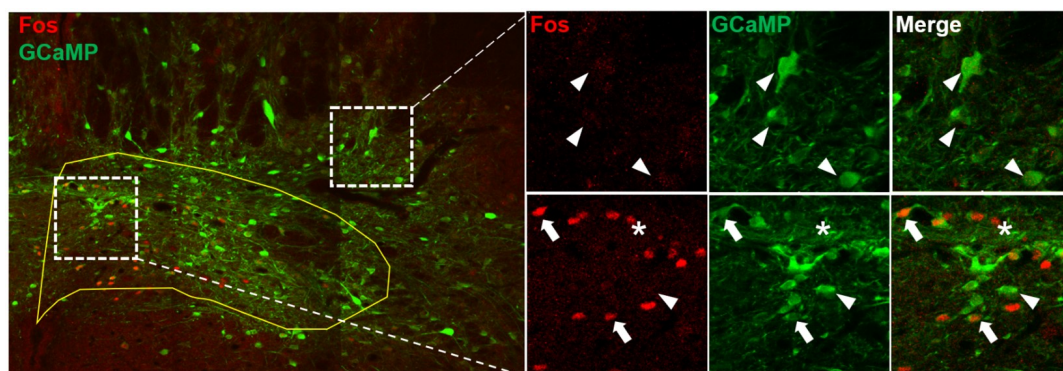
to alterations in neural activity in the nTS, be explained by altered activity of neurons located outside the nTS?

The reviewer is correct that our viral spread extends beyond the boundaries of the nTS, raising the possibility that the responses observed in Figure 5 may be influenced by neural activity of cells outside the nTS. While some viral spread beyond the target region is unavoidable, calcium transients were measured at the tip of the fiber, which was positioned directly within the nTS.

To address this concern further, we performed Fos immunohistochemistry in a subset of animals that received bilateral GCaMP virus injections into the nTS. Following fentanyl administration (50 µg/kg IV), brains were collected two hours later. As shown in the accompanying image, we observed Fos-IR co-expression with GCaMP exclusively within the nTS boundaries. No Fos-IR was detected outside the nTS, including in GCaMP cells. Taken together, these findings support our conclusion that the data depicted in our photometry figure (now Figure 6) accurately represent fentanyl-induced activity changes in nTS neurons.

Author response image 1.

Arrowheads: Fos-negative GCaMP cell; Arrows: Co-labeled Fos/GCaMP cell; Asterisk: Fos+ GCaMP-negative cell



(6) Currently, the *cFos* and photometry data are descriptive in nature. Are opioid-induced changes in nTS neural activity relevant to respiratory depression? If so, one might expect DREADD-mediated stimulation of the nTS neural activity (or stimulating nTS activity by some other means) would reverse fentanyl-induced respiratory depression similar to naloxone and methyl-naloxone.

The reviewer raises an interesting point regarding the relevance of the nTS in the context of OIRD. The nTS is a major site of integration of sensory afferent information and involved in the initiation of reflex responses that facilitate a return to homeostasis. As described above, we characterized the collective response of nTS neurons to intravenous fentanyl using both Fos immunohistochemistry and fiber photometry. Our data indicate that fentanyl-induced changes in nTS activity are strongly mediated by peripheral MOR. While the suggestion to use global chemogenetic activation of nTS neurons to reverse fentanyl-induced respiratory depression is intriguing, results from these experiments may be difficult to interpret due to the extensive heterogeneity of the nTS. However, we are currently conducting similar experiments using a more selective approach that will allow us to isolate and evaluate specific nTS phenotypes to better understand their contributions to OIRD.

(7) Are peripherally restricted mu opioid receptor (MOR) agonists available? If so, it would strengthen the paper if such compounds could be used to show that stimulation of

peripheral MORs is sufficient to induce respiratory distress independent of actions on centrally located MORs.

Peripherally acting Mu Opioid Receptor Antagonists (PAMORAs) are indeed available and currently being evaluated in our laboratory.

Reviewer #2 (Recommendations for the authors):

Consider having the figures/data numbered in the order that they appear in the manuscript. Right now, Figure 6 is mentioned between Figures 1 and 2 (minor).

Thank you for this suggestion. We have reordered the figures so that the biodistribution figure appears before the MOR antagonist pretreatment and reversal figures.

Reviewer #3 (Recommendations for the authors):

This manuscript outlines a series of very exciting and game-changing experiments examining the role of peripheral MORs in OIRD. The authors outline experiments that demonstrate a peripherally restricted MOR antagonist (NLX Methiodide) can rescue fentanyl-induced respiratory depression and this effect coincides with a lack of conditioned place aversion. This approach would be a massive boon to the OUD community, as there are a multitude of clinical reports showing that naloxone rescue post fentanyl over-intoxication is more aversive than the potential loss-of-life to the individuals involved. This important study reframes our understanding of successful overdose rescue with potential for reduced aversive withdrawal effects.

While this is an exciting and important study, there are a few minor to moderate critiques for the authors to consider. These are below.

(1) Title: "devoid of aversive effects" - While CPA is a good, cumulative indicator of potential aversive effects, it is not an exhaustive one. Since no other withdrawal measures were included, this is an overstatement.

The reviewer is correct in noting that our analysis of aversive effects is not exhaustive. Since we only assessed changes in aversive behavior between NLX and NLXM, we believe it is more accurate to modify the title accordingly. We have changed the title from “devoid of aversive effects” to “devoid of aversive behavior” better reflect the scope of the experiments conducted.

(2) Page 3, top line: MOR (mu opioid receptor) is highly expressed...

An article should likely be included prior to MOR or make plural and adjust the sentence.

Thank you for this suggestion. We have reworked this section in the manuscript.

(3) Figure 6D: this figure is very important for the interpretation of every single figure. It should either be moved to figure 1 or 2 or combined with figure 1 or 2.

Thank you for this suggestion. The biodistribution figure has been moved to Figure 2.

(4) Page 5, line 164, Figure 21-D: remove the 1.

Done.

(5) Sex differences (or lack thereof):

Throughout the manuscript, the authors report a lack of sex differences. However, while the data is not powered for the distinction of sex differences, there appears to be a bimodal distribution of the individual data points that likely correspond to sex across most experiments. For example, in Figure 2E there are both color and clear dots, which this reviewer assumes indicates sex (however, this wasn't easily apparent if it was commented on at all in the paper). If you look at the saline oxygen saturation (nadir) levels (2e), there is wide variability with the red-filled circles, but not the clear ones. This may indicate a bimodal distribution (and may be related to the baseline HR sex differences highlighted). This is also the case in Figure 2L but is perhaps more obvious in the CPA score data (Figure 4d), where it seems the nlx negative CPA effects were likely driven primarily by one sex. While this reviewer does not expect a full powering of experiments for sex differences (and also is very appreciative of the inclusion of both sexes), full raw data with sex indicated included in the supplemental data would greatly aid the field in general and allow for those with a specific interest in this area to build upon this data. Additionally, further discussion regarding the potential role of sex differences in the translational value of these findings is also warranted.

For all bar graphs, open symbols represent females and filled symbols represent males. This information can be found in the first paragraph of the Materials and Methods section. We have also added this information to each figure for increased visibility. We appreciate the acknowledgement of our inclusion of both sexes. For all experiments, we attempted to balance by sex. Unfortunately, we occasionally had to exclude animals for technical reasons (with clogged catheters being the most common reason for exclusion). This sometimes led to an imbalance in sex in some groups, as the reviewer has noted. In the graph of oxygen saturation nadir values in Fig 2E (now Fig 3E in the revised manuscript, all animals received intravenous fentanyl at a dose of 20 ug/kg. The reviewer is correct that there is greater variability in the males (filled symbols) compared to the females (open symbols) in this graph. However, this variability in the distribution was not observed in Fig 1E or Fig 4E, in which male and female rats received an identical dose of 20 ug/kg. Taking this into account, our overall interpretation of the data is that there is relatively minor sex difference in the responses observed after intravenous fentanyl, and the variability in Fig 3E is primarily due to a lower n compared to Fig 1E.

All raw data will be uploaded to a data repository.

(6) Page 7, line 209: Figure 5D should be Figure 6D.

We have incorporated this change.

(7) Page 8, line 267: Cure should be Curve.

We have incorporated this change.

(8) Discussion: Page10, line322 states that "no detectable NLX ... was found in brain tissue". This is incorrect based on Figure 6.

The sentence the reviewer highlighted refers to detection of NLX or NLXM in brain tissue from animals that received intravenous NLXM. As demonstrated in the biodistribution figure (now Figure 2 in the manuscript), our data demonstrate that an intravenous injection of NLXM did not result in NLX formation in the brain. We have reworked the sentence for clarity.

(9) jGCaMP injections: Figure 5B/c shows the distribution of the gcamp across animals. The optic fiber is placed directly over the NTs. However, how are we certain there isn't a

nearby nuclei/structure outside the NTS that is contributing to the photometry data presented in D-G?

See our above comment.

(10) *Fiber Photometry and Sex: These studies unfortunately may have had only 1 of a sex included in the fiber photometry data. While the inclusion is overall good, the single value for a sex suggests that there are differences, given the clustering of the data. While the anesthesia may be driving this potential sex effect, it is not clear based on the data presented. For reference: <https://link.springer.com/article/10.1007/s12975-012-0229-y>*

The reviewer is correct that there was an imbalance of sex in this dataset. While we made every attempt to balance for sex across all experiments, we unfortunately had to exclude some animals for technical reasons (clogged catheter, missed injection site, etc). This produced an imbalance in our photometry studies and did not allow us to thoroughly evaluate sex differences in fentanyl-induced changes in neural activity or in the responses to anesthesia. We have expanded on this limitation in the discussion.

(11) *Figure 5 - the bars are not the color indicated by the legend.*

We have corrected this in the figure. Thank you.

<https://doi.org/10.7554/eLife.104469.2.sa0>